ANTIBIOTIC RESISTANCE

Tuberculosis treatment failure associated with evolution of antibiotic resilience

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The widespread use of antibiotics has placed bacterial pathogens under intense pressure to evolve new survival mechanisms. Genomic analysis of 51,229 *Mycobacterium tuberculosis* (*Mtb*) clinical isolates has identified an essential transcriptional regulator, *Rv1830*, herein called *resR* for resilience regulator, as a frequent target of positive (adaptive) selection. *resR* mutants do not show canonical drug resistance or drug tolerance but instead shorten the post-antibiotic effect, meaning that they enable *Mtb* to resume growth after drug exposure substantially faster than wild-type strains. We refer to this phenotype as antibiotic resilience. ResR acts in a regulatory cascade with other transcription factors controlling cell growth and division, which are also under positive selection in clinical isolates of *Mtb*. Mutations of these genes are associated with treatment failure and the acquisition of canonical drug resistance.

Antibiotics are designed to kill bacteria by inhibiting biological processes essential for their survival. Completely eliminating a population of bacteria with antibiotics is challenging even when the bacteria are nominally sensitive to the drug (*1*). There is a growing body of work defining the mechanisms beyond canonical resistance by which bacterial populations avoid clearance by antibiotics (*2*, *3*). These include the formation of privileged subpopulations of drug-tolerant bacteria, as well as the acquisition of other traits that enhance bacterial survival, including variations in the cellular requirements for the target and alterations in drug penetration or efflux (*2*–*4*). We hypothesized that bacterial adaptation to antibiotics would be recorded as mutations in the genes associated with the most clinically relevant of these adaptive processes, and determining the targets of natural selection would reveal new mechanisms that enable bacteria to survive antibiotics in patients. Therefore, we sought to identify the evolutionary signatures of adaptation to antibiotics in *Mycobacterium tuberculosis* (*Mtb*). This obligate human pathogen has remained one of the largest causes of mortality despite facing months of antibiotic pressure in every treated patient and having been under antibiotic selection globally since the introduction of streptomycin in 1944 (*5*).

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Ongoing positive selection in the *Mtb* population

When clinical *Mtb* isolates are sequenced, the specimen is typically not derived from a single colony, but rather represents the bacterial population sampled from the patient. This population contains mutations arising from within-host evolution of the bacterium, which can be tracked by identifying unfixed mutations detected in deep sequencing (Fig. 1A) (*6*, *7*). We reasoned that by looking at within-host evolution across very large numbers of clinical isolates, even in the absence of clinical metadata, we could identify genes that repeatedly acquire mutations (parallel evolution) within different patients that would mark processes under positive selection. Therefore, we assembled previously published whole-genome sequencing data from 51,229 *Mtb* clinical isolates (Fig. 1B) and used a mutation burden test and the ratio of nonsynonymous to synonymous mutations (*dN/dS*) to identify genes that are under positive selection in the *Mtb* population (see the materials and methods). We also identified intergenic regions (IGRs) that are highly enriched for mutations. As expected, genes and IGRs previously associated with drug resistance or tolerance were highly mutated and the genes had *dN/dS* ratios >1, which is indicative of positive selection (Fig. 1B and fig. S1). We also identified a second set of genes that showed similar selective signals but for which the selective pressures are unknown (Fig. 1B).

Functional enrichment analysis indicated that the genes under selection are highly enriched for transcriptional regulators (12/60, $P = 1.57 \times 10^{-5}$; Fig. 1, B and C). One of the most frequently mutated genes is *resR* (gene identifier: *Rv1830*), an essential gene that is predicted to be a *merR*-type regulator. We also found evidence of positive selection in an analysis of the fixed mutations in *resR* (*dN/dS* = 2.91). The pattern of mutation was very similar between unfixed and fixed mutations (fig. S2), suggesting that within-host selection leads to mutation fixation. The *dN/dS* of *resR* was higher in drug-resistant strains compared with drug-sensitive strains (Fig. 1D), suggesting that *resR* mutations are associated with the evolution of drug resistance and consistent with a recent study showing that *resR* mutations were associated with a multidrug-resistant tuberculosis (TB) phenotype (*8*). Protein structural modeling indicated that most *resR* clinical mutations occur in the *merR*-type helix-turn-helix DNA-binding domain, with the most common mutations located at the interface between ResR and duplex DNA (Fig. 1E and fig. S2), where they could affect DNA-binding affinity and the transcription of downstream genes.

resR mutants display antibiotic resilience

To investigate the functional consequence of *resR* mutations, we used oligo-mediated recombineering to make point mutant strains by individually introducing three common clinical mutations (P59L, A85V, and R95C) into the chromosomal copy of *resR* in *Mtb* (H37Rv) (Fig. 1F). These *resR* mutations did not affect bacterial growth under standard or host-relevant carbon source conditions (fig. S3, A and B). Given the association with drug resistance, which occurs through de novo chromosomal mutation in *Mtb*, we also assessed the effect of *resR* mutations on the bacterial mutation rate, but did not find alterations (fig. S3C and table S1). However, we noted that *resR* mutants were ~20% longer and ~5% wider than the wild-type cells (fig. S4) and had a thickened cell envelope (fig. S5), suggesting that mutations of *resR* have functional consequences on bacterial size control under standard growth conditions.

We next tested the effects of *resR* mutations on drug resistance by measuring the minimum inhibitory concentration (MIC) for a panel of eight antibiotics, including first-line, second-line, and new anti-TB drugs (Fig. 2A). *resR* mutants and wild-type strains had very similar MIC profiles, with only subtle MIC shifts identified: a small increase for isoniazid (0.01 to 0.02 μg/ml) and minor decreases for rifampicin and ofloxacin (Fig. 2A and fig. S6). These changes are far below the MIC designated for being classified as drug resistant (Fig. 2A) (*9*). We next assessed the effects of *resR* mutations on drug tolerance by performing time-kill analyses for the eight antibiotics using drug concentrations 100-fold MIC. The *resR* mutants and wild-type strains showed very similar time-kill dynamics, although again, there were subtle differences for some drugs (ofloxacin,

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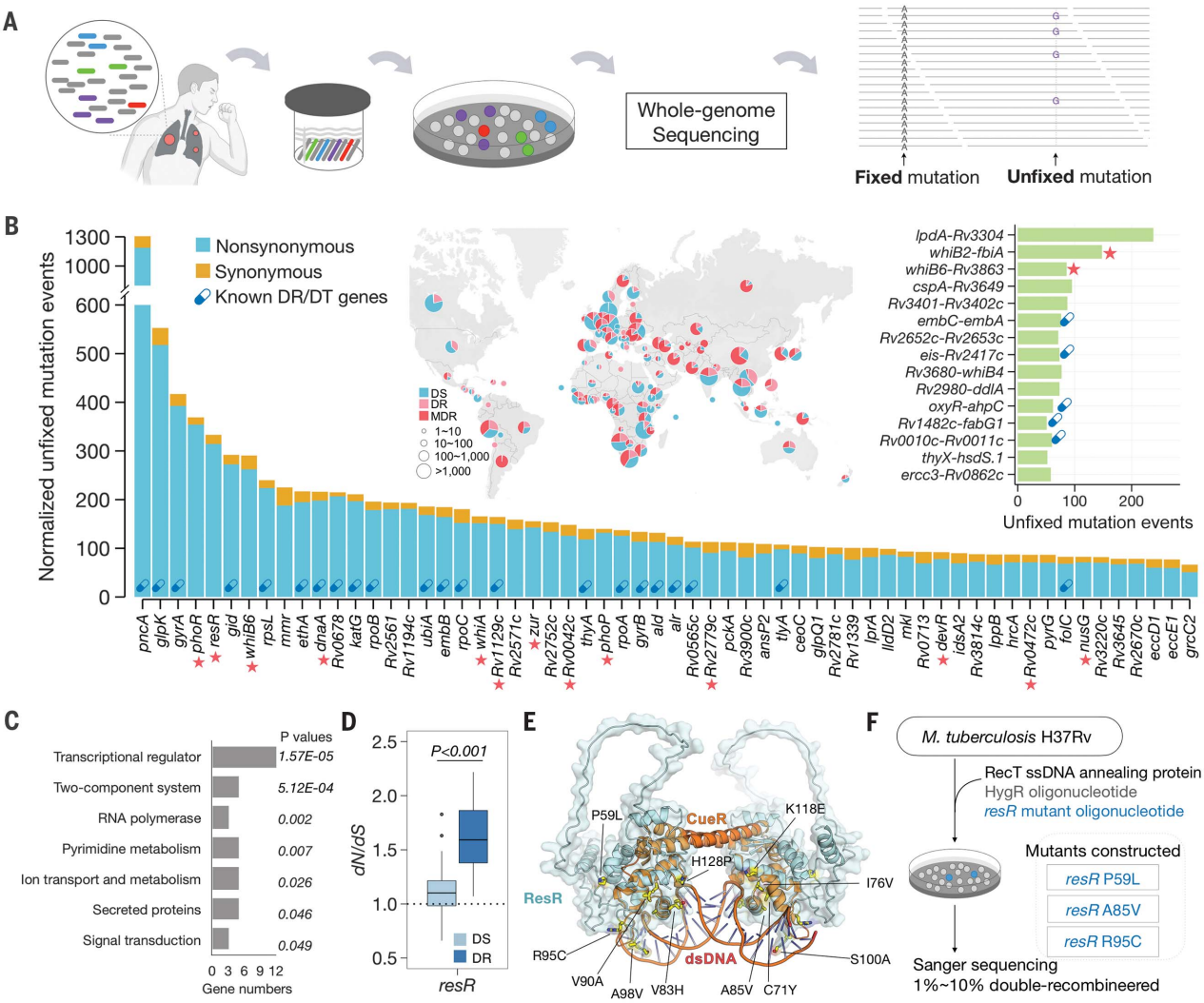

Fig. 1. Ongoing positive selection in the global *Mtb* population. (**A**) Diagram of fixed and unfixed variants. (**B**) Genes and IGRs with signal of ongoing positive selection. Known drug-resistance (DR) or drug-tolerant (DT) genes are marked with capsule symbols. Red stars indicate transcriptional regulators. The map shows the geographic origin of the clinical *Mtb* isolates and their DR status. MDR, multidrug resistance. (**C**) Categories of genes under positive selection with adjusted *P* values for enrichment. (**D**) *dN/dS* ratio of *resR* in DR and DS strains. *P* value was determined by unpaired *t* test. (**E**) Structural model of the ResR dimer (cyan, predicted by Alphafold) aligned to the MerR-family homolog CueR (orange) in complex with duplex DNA with common ResR clinical mutations indicated (yellow). (**F**) Schematic of construction of mutations in the chromosomal copy of *resR* in *Mtb*.

ethionamide, and bedaquiline) at particular time points (Fig. 2B). The changes in the minimum duration of time to kill 99% of bacteria (MDK_{99}) were <24 hours (Fig. 2B); previously characterized drug-tolerance mutants in *Mtb* prolonged the MDK_{99} by more than 4 days (*10*).

In contrast to these modest changes in MIC and MDK_{99}, in the time-kill assays, we observed a strong and unexpected drug phenotype that clearly distinguished *resR* mutants from the wild-type strains. Once the cells were plated on drug-free media after antibiotic exposure, *resR* mutants formed visible colonies significantly earlier than the wild type (Fig. 2C). This early recovery phenotype of *resR* mutants was observed for all eight antibiotics tested, whereas no difference in growth was seen in the absence of drug exposure (Fig. 2C and fig. S7). Using quantitative image analysis to track the growth of individual colonies over time (Fig. 3A and fig. S8) (*11*), we found that antibiotic treatment caused a 4.7- to 6.8-day delay in colony formation for wild-type *Mtb*; these delays were highly reproducible and characteristic for each drug (Fig. 3B). The *resR* mutants reduced the time to colony formation by 20 to 50% for all eight drugs tested (Fig. 3, B and C). The *resR* mutants also showed faster recovery when challenged by combinations of the first-line anti-TB drugs (fig. S9, A to C). We calculated the growth dynamics of traced colonies over time and found that the expansion rate of visible colonies did not differ between WT and mutant cells (fig. S9D), suggesting that faster recovery occurred at early stages before colony appearance.

A delay in bacterial growth after a period of antibiotic exposure is classically referred to as the post-antibiotic effect (PAE) (*12*). The PAE has been recognized for decades across many bacterial pathogens and is an important factor in the design of treatment regimens (*12*, *13*). To date, the PAE remains mechanistically enigmatic, with recent work favoring a detoxification model in which the cells must export residual drug before growth resumes (*14*). Our results suggest that *Mtb* has mechanisms for post-antibiotic recovery that are under genetic control, and that *resR* mutants optimize this system, accelerating bacterial recovery after a wide variety of antibiotic insults. We

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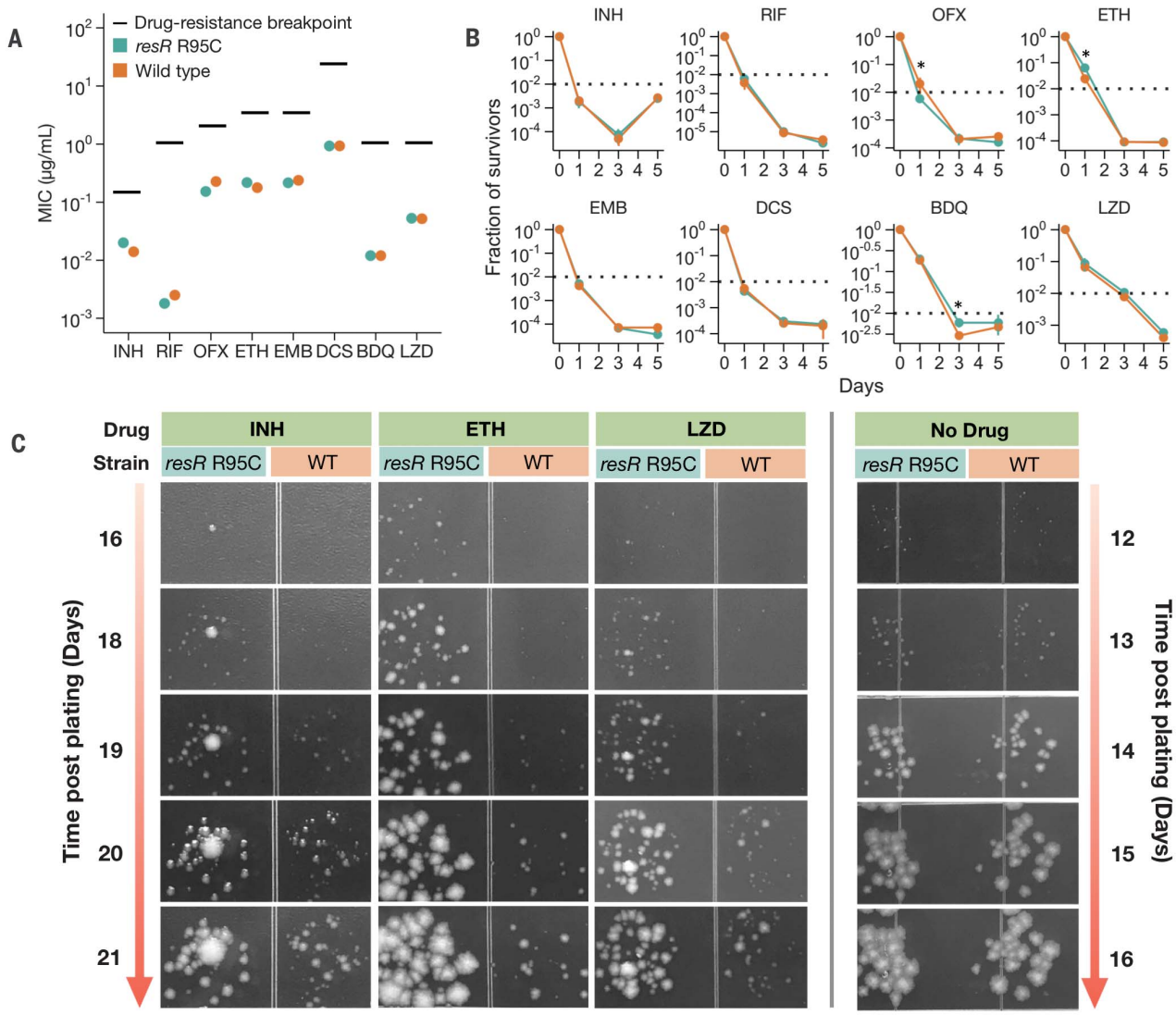

Fig. 2. *resR* mutants show faster recovery after drug exposure. (**A**) MICs of *resR* mutants and wild type for eight anti-TB drugs: INH, isoniazid; RIF, rifampicin; OFX, ofloxacin; ETH, ethionamide; EMB, ethambutol; DCS, D-cycloserine; BDQ, bedaquiline; LZD, linezolid. The drug-resistance breakpoint is the drug concentration that defines clinical drug resistance. (**B**) Time-kill kinetics for eight different drugs for *resR* mutant and wild-type *Mtb* strains. The concentration of each drug was 100-fold MIC. The dashed line indicates MDK_{99}. *$P < 0.05$ by unpaired *t* test. (**C**) Representative images illustrating the post-antibiotic recovery dynamics of *resR* mutant (R95C) and wild-type (WT) *Mtb* strains.

refer to this phenotype as antibiotic resilience and thus named this gene *Rv1830*, resR for resilience regulator.

Antibiotics resilience manifests as early recovery

To better understand early events in post-antibiotic recovery, we used microscopy to investigate cell regrowth after the period of antibiotic exposure. *Mtb* grow from their subpolar regions, which can be quantitatively defined by imaging that tracks the incorporation of fluorescently labeled D-amino acids (FDAAs) into the nascent peptidoglycan near the cell poles (*15*, *16*). We assessed regrowth of wild-type and *resR* mutant cells after 24 hours of isoniazid exposure at 100-fold MIC, followed by a 10- to 36-hour recovery window in drug-free media (Fig. 4, A and B). During the recovery period, *resR* mutants showed significantly more new cell wall synthesis than the wild-type cells ($P < 0.0001$ for 24 and 36 hours; Fig. 4, B and C). Wild-type cells and *resR* mutants had similar FDAA incorporation in the absence of antibiotic exposure (Fig. 4B).

This labeling approach allowed us to use flow cytometry to assess new cell wall synthesis rapidly and quantitatively in mutant and wild-type cells across a wide range of antibiotic exposure conditions. We assessed regrowth after exposure to eight antibiotics at 100-fold MIC. *resR* mutants showed increased cell wall synthesis after exposure to all antibiotics tested, whereas label incorporation was similar between *resR* mutants and the wild type in the absence of drug (Fig. 4D). The different *resR* mutants all showed a similar phenotype (fig. S9E). *resR* mutants also demonstrated faster recovery after up to 120 hours of isoniazid exposure and across a range of antibiotic concentrations (fig. S10).

A regulatory pathway underlying antibiotic resilience

To delineate the regulatory targets through which ResR mediates antibiotic resilience, we used an in vitro whole-genome DNA binding and deep sequencing assay (IDAP-Seq) (*17*) to identify the ResR-binding sites across the genome. We purified ResR proteins with either an N-terminal or a C-terminal His tag, and

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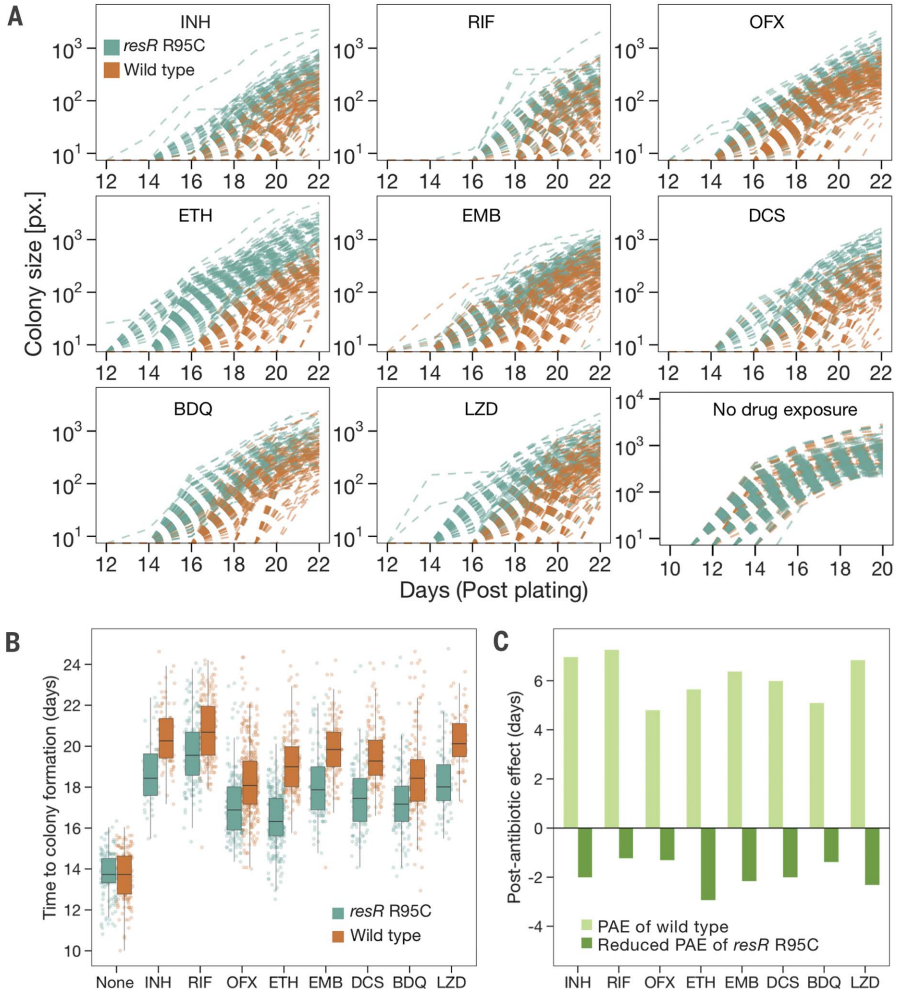

Fig. 3. Quantitative colony size tracking indicates shortened PAE in *resR* mutants. (**A**) Quantitative colony size tracking in pixels (px.) for *resR* mutant and wild-type *Mtb* strains in the presence or absence of antibiotic exposure for 24 hours (24h). (**B**) Duration from plating to the appearance of visible colonies (time to colony formation) for *resR* mutant and wild-type strains after exposure to the indicated antibiotics (24h) or no drug exposure (none). $P = 0.7291$ for the no-drug-exposure group and $P < 0.0001$ for all drug groups, as determined by Mann-Whitney U test. (**C**) Bar plot depicting the median values of post-antibiotic delay of wild-type (light green) and *resR* mutant strains (dark green).

performed IDAP-Seq with both versions. With both constructs, we identified four major ResR-binding sites in the *Mtb* genome in the intergenic regions between *rimJ-Rv0996*, *nrdH-Rv3054*, *whiB2-fbiA*, and *Rv3916c-parB* (Fig. 5A). None of the genes flanking the dominant ResR binding sites has a known role in drug resistance. Unexpectedly, however, our population genomic analyses found that one of them, the *whiB2-fbiA* intergenic region, is also one of the major targets of ongoing positive selection in clinical *Mtb* isolates (Fig. 1B). Indeed, the ResR-binding site in the *whiB2-fbiA* intergenic region is the site of mutant accumulation in clinical *Mtb* strains (Fig. 5B), an evolutionary pattern suggesting that natural selection has acted on both the ResR and its binding site, presumably to alter the expression of *whiB2* or *fbiA*.

To assess the regulatory role of ResR, we used RNA sequencing to assess gene expression changes in the three independent *resR* mutants compared with isogenic wild-type *Mtb*. The *resR* mutants had ~2-fold higher levels of *whiB2* transcript but no significant alterations in the expression of the other genes adjacent to the major ResR-binding sites (Fig. 5C). The changes in *whiB2* expression were part of a pattern of altered transcription that also included increased expression of the *iniBAC* operon, which was previously shown to be expressed in response to cell wall–targeting antibiotics (*18*) and other mediators or regulators of central carbon metabolism (e.g., *bkdABC* and *clgR*) and cell wall–acting genes, notably *pbpB* (*19*, *20*) (fig. S11A). To further define the regulatory relationship between *resR* and *whiB2*, we reconstructed several *resR* variants in *Mycobacterium smegmatis* (homolog: *MSMEG_3644*), including CRISPR-i knockdown, overexpression, and point-mutant strains (Fig. 5D). Transcriptional profiling of these strains suggested that ResR functions as an activator of *whiB2* expression, and that the clinical mutation T77A results in increased *whiB2* transcription (Fig. 5D and fig. S11B).

WhiB2 is an essential regulator that controls cell growth division in mycobacteria. Knockdown of *whiB2* expression results in cell elongation and division defects, whereas overexpression of *whiB2* in *Mtb* results in increased cell length (*21*, *22*). We next sought to determine the effect of the *whiB2-fbiA* mutation on *whiB2* expression and *Mtb* physiology. We generated isogenic *Mtb* strains carrying a clinically prevalent mutation in *whiB2-fbiA* (3640375 C>T). The point-mutant strain had an ~1.6-fold increase in *whiB2* expression (Fig. 5E and fig. S12A). Moreover, this strain also phenocopied *resR* mutants, with similar morphotypic changes (longer and wider cells) (fig. S12B), a small increase of isoniazid MIC, and, most importantly (fig. S12C), faster recovery after antibiotic exposure (fig. S12, D and E). However, the magnitude of the resilience phenotype in the *whiB2-fbiA* mutant appeared to be smaller than that of *resR* mutants, suggesting that additional, as yet undefined, mechanisms might be involved.

resR, *whiB2-fbiA*, and *whiA* mutants are associated with acquisition of canonical drug resistance and treatment failure

The WhiB2 regulon has been best characterized in *Streptomyces*, in which the WhiB2 homolog

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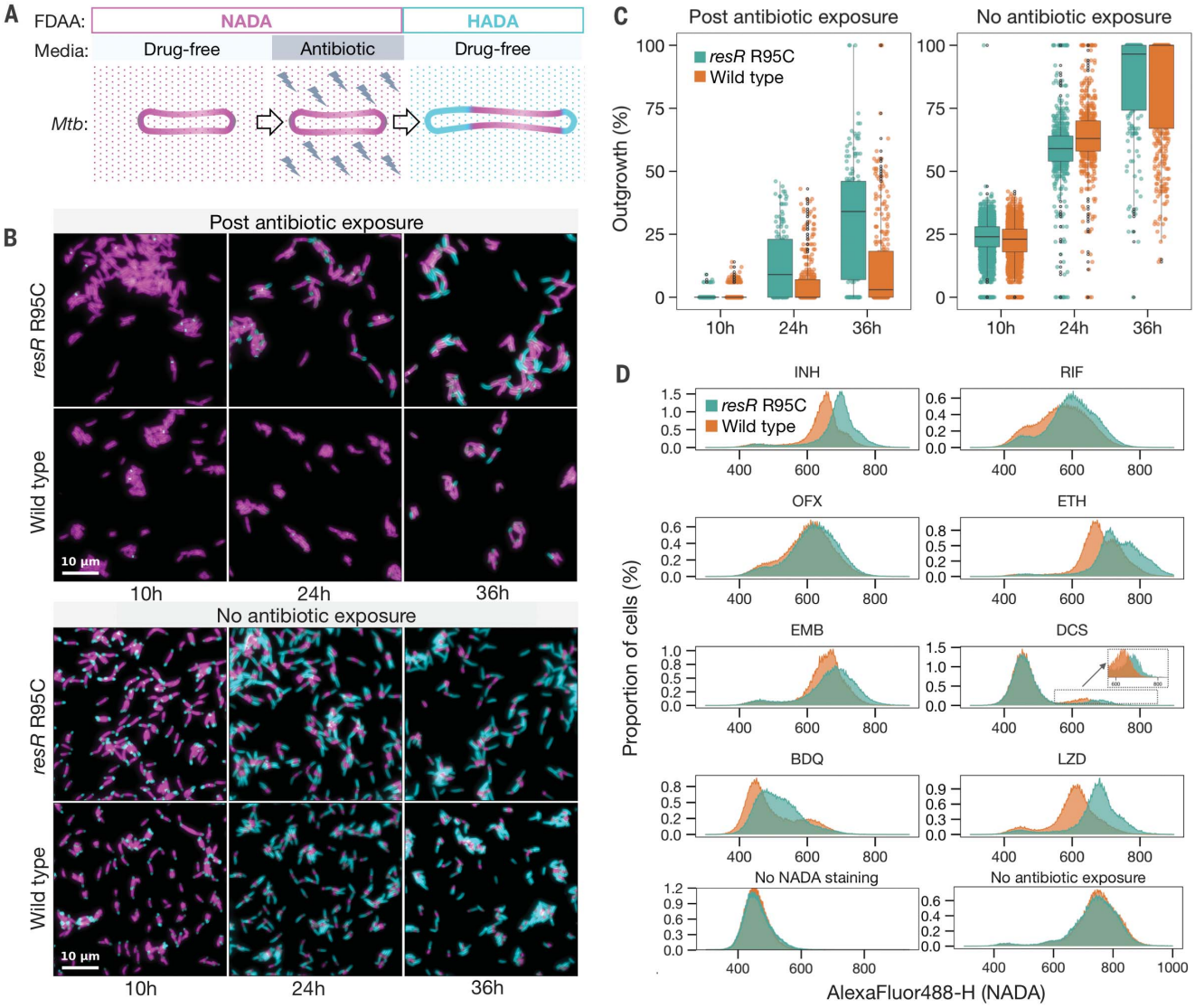

Fig. 4. Antibiotic resilience characterized by more rapid resumption of cell wall synthesis. (**A**) Schematic diagram of a pulse-chase experiment for tracking the regrowth of *Mtb* cells after antibiotic exposure using FDAA incorporation (NADA and HADA). (**B**) Representative microscopy snapshots of the *resR* mutants and wild type during regrowth after 24 hours of INH (100-fold MIC) exposure (top) and no antibiotic exposure (bottom). (**C**) Quantitative comparison of outgrowth between *resR* mutant and wild type in the post-antibiotic-exposure group and the no-antibiotic-exposure group; $P < 0.0001$ for 24 and 36 hours of post-antibiotic-exposure group by double-sided Kolmogorov–Smirnov test. (**D**) Flow cytometry for NADA incorporation into *Mtb* cells after 24 hours of recovery with or without antibiotic exposure.

functions with another regulator, WhiA, to initiate cell division (*23*, *24*). We found in our population genomic analysis that *whiA*, like *resR* and *whiB2*, is also a major target of positive selection in clinical *Mtb* isolates (Fig. 1B). To assess the clinical relevance of mutations in this regulatory triad, we compared the prevalence of fixed mutations of *resR*, *whiB2-fbiA*, and *whiA* in drug-susceptible strains versus drug-resistant strains. Of the 51,229 genomes in our database, 6.9% had mutations in these genes or IGRs, and mutations in these genes were significantly enriched in drug-resistant strains (fig. S13A). We further looked at *Mtb* strains from two high-TB-burden countries, India and China, where *Mtb* populations were not skewed by recent expansions of clonal *Mtb* strains, and each had >1000 sequenced isolates. Again, strains with mutations in the three genes or IGR had increased odds ratios (ORs) for drug resistance: India, OR: 1.80; 95% confidence interval (CI): 1.11~2.91; China, OR: 1.71; 95% CI: 1.06~2.74 (Fig. 6A and fig. S13B). Therefore, these data suggest that *resR*, *whiB2-fbiA*, and *whiA* mutations facilitate the evolution of canonical drug resistance. These mutations remain under positive selection in highly drug-resistant strains, as assessed by within-host evolution (Fig. 6B), suggesting that even after the emergence of resistance to first-line drugs, they provide an advantage to the bacterium as it encounters second-line agents. Evidence of selection at every level of drug resistance further supports the model that these variants are selected by a multidrug phenotype such as antibiotic resilience. However, the shifted isoniazid MIC is likely to contribute to their success in drug-susceptible strains, and this dual benefit may be one reason that they are clinically prevalent.

Finally, although *resR*, *whiB2-fbiA*, and *whiA* variants are more frequent in drug-resistant strains, variations in these genes are also under positive selection in drug-susceptible strains (fig. S13C), and the sites of mutation

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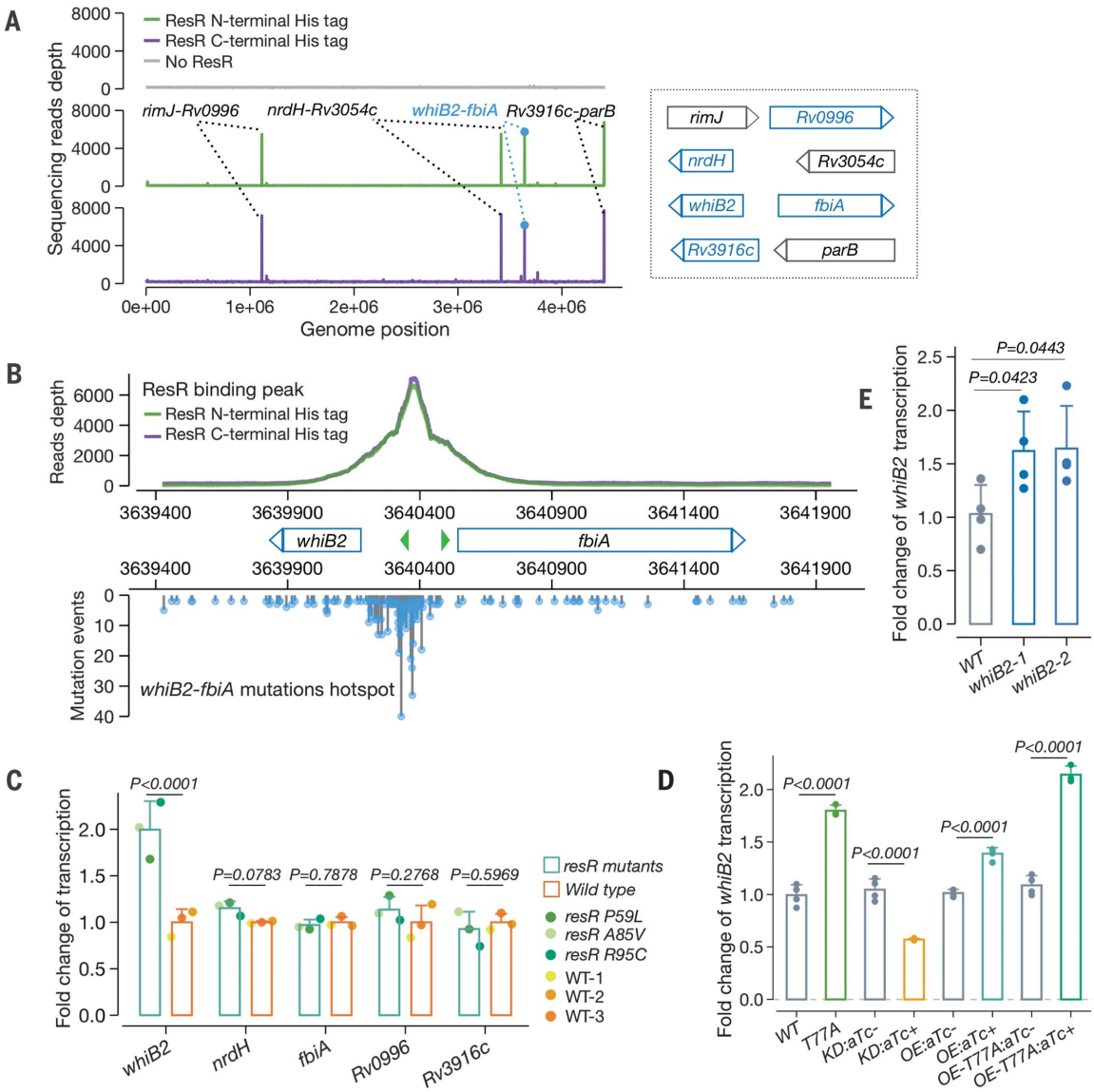

Fig. 5. ResR activates *whiB2* and clinical mutations leading to up-regulation of *whiB2*. (**A**) IDAP-Seq identified binding sites of ResR. (**B**) ResR-binding peaks overlaid with the mutations identified in clinical strains between *whiB2* and *fbiA*. The transcriptional start sites of *whiB2* and *fbiA* are indicated by green arrows. (**C**) Fold change of transcription of putative ResR targets in *resR* mutants and wild-type strains. *P* values were determined by the Wald test implemented in DEseq2. (**D**) Transcriptional changes of *whiB2* in *M. smegmatis* strains: wild-type (WT), *resR* point mutant (T77A), CRISPR-i knock-down of *resR* (KD), merodiploid overexpression of wild-type *resR* (OE) or T77A mutant form (OE-T77A). The absence or presence of aTc is indicated by (−) or (+). *P* values were determined by unpaired *t* test. (**E**) The *whiB2-fbiA* mutant (3640375 C>T) exhibited up-regulation of *whiB2*. *whiB2-1* and *whiB2-2* refer to the two parallel mutants. *P* values were determined by unpaired *t* test.

in drug-susceptible strains mimic those in drug-resistant strains (fig. S13D). Indeed, 1.5 to 9.7% of drug-susceptible *Mtb* strains from the high-TB-burden countries had fixed mutations in one of these three genes or IGRs (fig. S13E). We postulated that in addition to being clinically important as stepping-stone mutations for the emergence of canonical drug resistance, these variants independently contribute to treatment failure in drug-susceptible patients. To test this hypothesis, we reanalyzed data from the REMoxTB trial, a global phase 3 clinical trial that sought to reduce the treatment duration of drug-susceptible TB (*25*, *26*). Of the 1931 patients enrolled in the trial, paired *Mtb* isolates from 36 recurrent TB patients were whole-genome sequenced and the isolates remained drug sensitive (*26*). We found that eight of 36 (22.2%) isolates from patients who failed treatment had *resR*, *whiB2-fbiA*, or *whiA* mutations (Fig. 6C), a frequency significantly higher than found in the background populations of South Africa (108/2998, 3.6%) and Thailand (110/2036, 5.4%), where the trial was conducted (P = 0.0024 and P = 0.0152, respectively, Fisher's exact test; Fig. 6D). Additionally, the within-host frequency of these variants increased between the initial and recurrent episodes, in three cases rising to fixation (Fig. 6E).

We have identified a new form of altered drug susceptibility that we call antibiotic resilience, which is regulated by the gene *resR*. Mutations in the regulating cascade are positively selected in clinical *Mtb* isolates and associated with poor treatment outcomes in TB. Current

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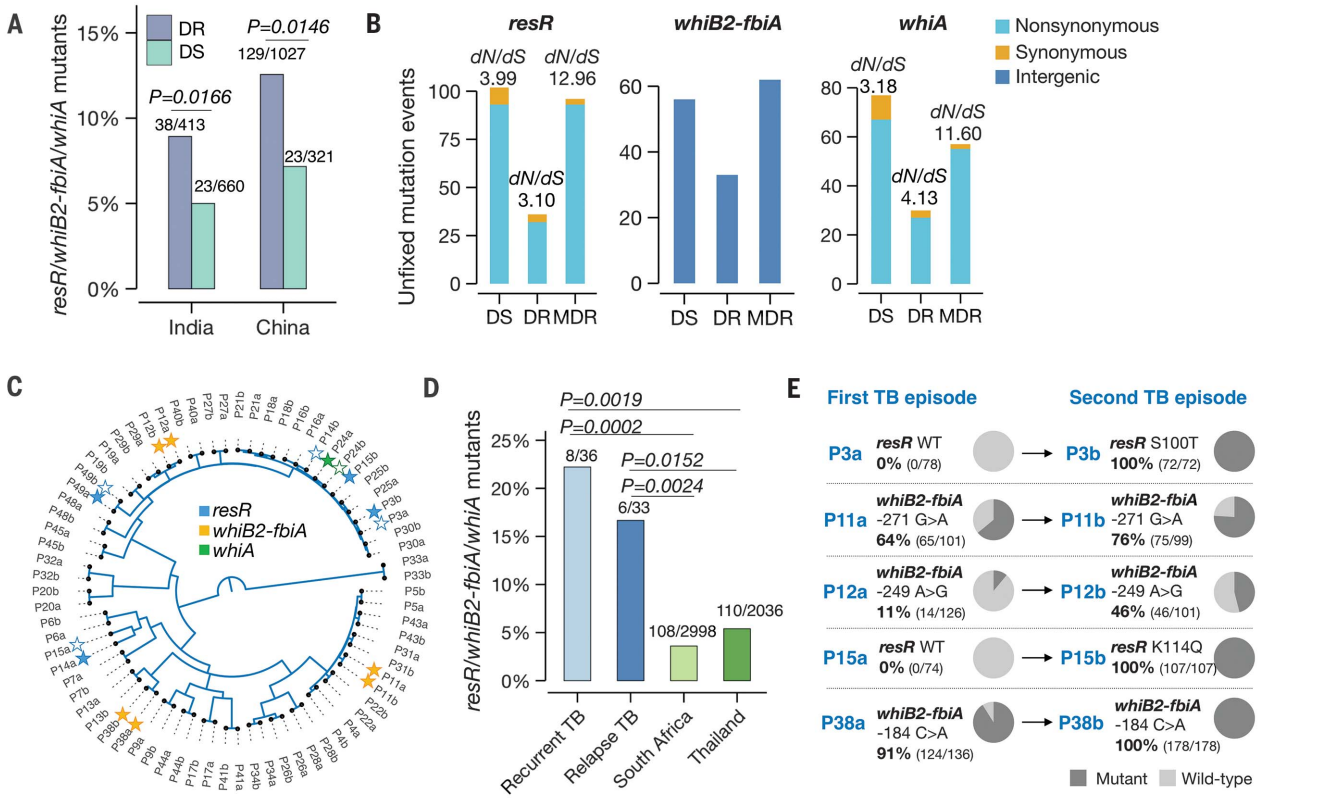

Fig. 6. *resR*, *whiB2-fbiA*, and *whiA* mutants are associated with canonical drug resistance and relapse of drug-susceptible TB. (**A**) Proportion of *resR*, *whiB2-fbiA*, and *whiA* mutants in DR and DS strains sequenced from India and China. *P* values were determined by Fisher's exact test. (**B**) Unfixed mutations in DS, DR, and MDR (resistant to RIF and INH) strains. (**C**) Phylogenetic tree of paired *Mtb* isolates from 36 recurrent TB patients. Solid stars indicate isolates in which mutations were detected, and empty stars indicate the absence of mutations in one of the paired isolates. (**D**) Percentage of isolates with mutations in *resR/whiB2-fbiA/whiA*. "Recurrent TB" includes three patients with *Mtb* isolates suggestive of reinfection. (**E**) Mutational trajectory in *Mtb* isolates from the first TB to second TB episodes in five pairs of isolates. The mutations in other three pairs were P14 (*resR*, D144N; 100% in P14a, 0% in P14b), P49 (*resR*, R95H; 8.7% in P49a, 0% in P49b), and P24 (*whiA*, A131T; 11.6% in P24a, 0% in P24b).

clinical measures of antibiotic susceptibility focus largely on the magnitude of drug exposure; that is, what concentration of drug the pathogen can experience and grow or at least survive (*27*). However, our results suggest that in the clinical setting, *Mtb* is evolving to change the temporal dynamics of its recovery after drug exposure. The temporal dynamics of drug responses are considered when developing new drugs and drug regimens but are often forgotten as potential drivers of treatment outcomes that might be influenced by pathogen variation. In studies of *Escherichia coli* experimentally evolved under intermittent drug exposure, temporal variants were the first to emerge (*28*). Further, the mutants had delayed regrowth rather than faster regrowth, suggesting that the temporal phenotypes could vary by the pathogen, drug regimen, and host factors such as drug metabolism (*29–31*). Because current clinical diagnostics are largely blind to these phenotypes, selection for temporal phenotypes such as antibiotic resilience may underlie treatment failure in other drug-susceptible infections.

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SUPPLEMENTARY MATERIALS

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Materials and Methods
Figs. S1 to S13
Table S1
References (*33*–*61*)
MDAR Reproducibility Checklist

View/request a protocol for this paper from *Bio-protocol*.

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Tuberculosis treatment failure associated with evolution of antibiotic resilience

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Resilient in the long term

Tuberculosis caused by *Mycobacterium tuberculosis* is a persistent, sometimes lifelong infection that needs long courses of multiple antibiotics to treat. Not surprisingly, antibiotic resistance is rife. In a large sample of whole-genome data from clinical isolates, Liu *et al*. repeatedly observed an elongated phenotype showing rapid regrowth after typical regimens of antibiotic treatment. Signals of positive selection pointed to an essential transcriptional regulator (resR) and specific intergenic regions, which act in concert to regulate growth. Up to 10% of strains from high-tuberculosis-burden countries showed fixed mutations in these regions. Mutations across this cascade are associated with antibiotic treatment failure and are precursors to the emergence of classical antibiotic resistance. —CA

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